

# ngspice-46 — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

July 2026

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## ngspice-46 — Full Change Report

Every modification applied to ngspice-46 in this project, with its reason. The baseline is the pristine ngspice-46 source tree (as vendored in the project's `original/` snapshot, which already contains OpenVAF-Reloaded's stock OSDI support); the current state is the tree committed in this repository under `ngspice-46/`. Each entry links the enhancement write-up in [enhancements\\_doc/](#) that carries the full engineering detail and the verifying example suite. The companion document [openvaf\\_changes\\_full-report.md](#) covers the compiler side.

Maintenance note: this report is updated whenever an enhancement touches ngspice sources. If you are reading it alongside a newer tree, the per-enhancement index at the end tells you the last change it covers.

Scope summary. One new source file and 41 modified ones carry all the functional changes (~2,000 substantive diff lines). A further ~100 `Makefile.in` files and `config.h.in` differ only because the auto-tools were regenerated with a current libtool ([E-77](#)); they contain no hand-written changes and are not listed individually. Everything below is behavior, interface, or diagnostics.

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### 1. The OSDI ABI, version 0.4 → 0.7

`src/osdi/osdi.h` is the authoritative C contract between ngspice and the `.osdi` objects `openvaf-r` produces. The project extended it four times — twice additively (new exported symbols old loaders simply never look up), twice with genuine version bumps (layout changes):

| Change                                                                                                                       | Kind                                  | Enhancement          | Why                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>OSDI_ABSDELAY_COUNT</code><br>/<br><code>OSDI_ABSDELAY_INFO</code><br>exports                                          | additive<br>sym-<br>bols              | <a href="#">E-1</a>  | <code>absdelay()</code> needs simulator-side waveform history and synthetic delay-equation rows; the descriptors tell ngspice which nodes/offsets each delay slot uses                                                                                          |
| <code>OSDI_LAST_CROSSING_COUNT</code><br>/<br><code>OSDI_LAST_CROSSING_INFO</code><br>exports                                | additive<br>sym-<br>bols              | <a href="#">E-6</a>  | <code>last_crossing()</code> needs waveform history and crossing interpolation — same additive pattern as <code>absdelay</code>                                                                                                                                 |
| <code>OsdiNode.nodeset</code><br>field                                                                                       | ABI 0.5<br>(node<br>stride<br>change) | <a href="#">E-45</a> | Verilog-A net initializers ( <code>electrical a = 5.0;</code> ) become solver initial-guess nodesets; every node record carries the hint                                                                                                                        |
| <code>num_ac_stim_src</code><br>/<br><code>ac_stim_sources</code><br>/<br><code>load_ac_stim</code><br>descriptor tail       | ABI 0.6<br>(de-<br>scriptor<br>grows) | <a href="#">E-51</a> | full <code>ac_stim()</code> support: a Verilog-A module can <i>be</i> the AC stimulus, so the descriptor must enumerate its AC right-hand-side sources                                                                                                          |
| <code>load_noise</code><br>destination stride 1<br>→ 2 (signed<br>( <code>flat</code> , <code>react</code> )<br>power pairs) | ABI 0.7                               | <a href="#">E-54</a> | correct noise physics: operating-point-dependent factors and <code>ddt()</code> -shaped ( $j\omega$ ) noise need a complex amplitude per source, $re + j\omega \cdot im$ , so coherent same-named sources (LRM 4.6.4) sum with correct sign and frequency shape |

Two additive flag conventions ride on the existing `eval()` interface (not version bumps):

- `EVAL_FLAG_IS_FINAL_STEP` (bit 1 ≪ 21) in the `eval input` flags — `@(final_step)` firing ([E-53](#));
- `eval return` flags for `$finish/$stop` requests and `$discontinuity`-driven step rejection ([E-55](#)).

Pairing consequence: this ngspice requires  $\text{OSDI} \geq 0.7$  and rejects older objects with a clear version message; `.osdi` files from older compilers must be recompiled (E-54).

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## 2. The OSDI runtime — `src/osdi/`

**osdiaccept.c** — new file (E-55, E-1, E-6)

The accepted-timepoint hook (`DEVaccept`) OSDI previously lacked. It advances the `absdelay/last_crossing` waveform histories only at *accepted* points (so rejected timesteps don't corrupt the delay lines) and reads the per-attempt latched `$finish/$stop` request flags at the one boundary where honoring them is safe. Why: acting on `$stop` mid-Newton-iteration was treated as a step failure and ground the timestep into a rejection loop; `$finish` was ignored outright.

**osdidefs.h** (104 diff lines)

- `OsdiExtraInstData` grows the per-instance fields behind every runtime feature: `dt/temp` offsets with given-flags (instance temperature), `eval_flags`, `absdelay` waveform-history rows and pre-allocated KLU/SPARSE matrix-pointer sets for the delay-equation rows (re-pointed on every DC↔AC transition, mirroring the regular Jacobian handling), and the `last_crossing` history (E-1, E-6, E-55).
- `ALIGN` macro renamed `OSDI_ALIGN`: the macOS SDK's `<sys/param.h>` defines an unrelated `ALIGN`, producing a redefinition warning in every OSDI translation unit (E-77).

**osdiitf.h / osdiext.h**

The registry-entry structure gains the `absdelay/last_crossing` descriptor pointers, and `OSDIpendingRequests()` is exported so the analyses can ask "did any device request `$finish/$stop` at this accepted point?" (E-1, E-6, E-55).

**osdiregistry.c** (68 diff lines)

- Loads the `absdelay/last_crossing` descriptor arrays from each `.osdi` and threads them into the registry entries (E-1, E-6).
- Requires  $\text{OSDI} \geq 0.7$  (E-54).

**osdiinit.c** (8 diff lines)

- Registers the `DEVaccept` hook (E-55).
- Publishes the synthetic instance-temperature parameter under both spellings, `dt` and the conventional `dtemp` every built-in device uses; `n1 a b mod dtemp=10` used to fail with "*unknown parameter*" while the underlying plumbing was exact (E-80).

**osdiload.c** (441 diff lines — the largest OSDI change)

- Stamps the `absdelay` synthetic rows (`y_synth`, `z`) for DC, AC and transient, driving the delay from the recorded history (E-1).
- Evaluates `last_crossing` from the recorded waveform with linear interpolation between the bracketing timepoints (E-6).
- Passes the final-step flag through to `eval()` (E-53).
- Latches `eval`-return flags per timepoint attempt (`point_eval_flags`, OR-ed across Newton iterations, reset per attempt) so `$finish/$stop` under solution-dependent conditions survive to the accepted-point boundary (E-55).

- Folds the stride-2 signed noise-power pairs ( $\text{fac} \cdot |\text{fac}|$  semantics) when transferring `load_noise` results (E-54, E-42).

#### **osdisetup.c** (178 diff lines)

- Creates the synthetic delay-equation unknowns and matrix entries for `absdelay` (E-1).
- Applies the OSDI nodesets (`OsdiNode.nodeset`) as solver initial guesses (E-45).
- A `$fatal/$finish` raised during setup (models rejecting their configuration by design — HiSIM's `$port_connected` guards) used to surface as ngspice's baffling *"impossible error — can't occur"*; it now reads *"a Verilog-A device rejected its configuration during setup"* (E-56).
- An internal OSDI node that appears in no Jacobian entry — a net structurally decoupled from the matrix, e.g. an explicit ground `gnd` reference whose branch contribution `V(p,gnd) <+ ...` drops the `gnd` column — used to be allocated its own solver row. That all-zero row/column is benign under Sparse (it decouples to  $V=0$ ) but makes the KLU matrix structurally singular, so KLU returned a wrong DC answer (groundcontrib:  $v(p)=0$  not 1.5; hierbranch: branch currents 0). Such nodes are now tied to ground (node 0) at setup, matching how OpenVAF already treats them and fixing both solvers (E-116).

#### **osdiacld.c** (89 diff lines)

- AC loading gains the `ac_stim` right-hand-side injection: the partitioned AC-stimulus source array is loaded through `load_ac_stim` and added to the AC RHS with exact magnitude and phase, `$mfactor` applied linearly (deterministic signals scale with multiplicity) (E-51, E-26).
- Complex ( $j\omega$ -shaped) noise coupling entries participate in the AC matrix (E-54).

#### **osdinoise.c** (93 diff lines)

- Per-instance grouping of same-named noise sources: coherent summation of complex amplitudes  $(a + j\omega \cdot b) \cdot T$  before squaring, per LRM 4.6.4 — same-named sources correlate (including exact anti-phase cancellation), differently-named ones stay independent (E-42).
- Reads the stride-2 signed pairs of ABI 0.7 (E-54).

#### **osditrunc.c** (34 diff lines)

- Honors `$discontinuity(n)`'s next-step clamp: a negative `bound_step` sentinel from the model limits the step after the event (E-24).
- Honors the step-rejection return flag: when an event fired inside the step, `OSDitrunc` requests `delta/8` (with a  $20 \cdot \text{CKTdelmin}$  termination floor) so the integrator bisects onto the discontinuity instead of extrapolating across it (E-55).

#### **osdi/Makefile.am**

Adds `osdiaccept.c` to the build (E-55).

### 3. Analyses — **src/spicelib/analysis/**

#### **dctran.c** (46 diff lines)

- Advances OSDI delay/crossing histories via the accept path and skips history corruption on rejected steps (E-1, E-6).
- Issues the dedicated `@(final_step)` evaluation at the converged end of a successful transient (E-53).

- Honors accepted-point `$finish` (ends the analysis cleanly, firing `final_step` at the requesting point), `$stop` (pauses resumably), and aborts with a clear error on `$fatal`'s `E_PANIC` instead of retrying it as nonconvergence ([E-55](#)).

#### **dcop.c** (9) and **acan.c** (10)

- Fire `@(final_step)` at their successful end (an operating point fires both step events — a single point is first *and* last) ([E-53](#)).
- An AC job's operating-point phase carries the analysis name bit (only the name — adding the reactive `CALC_*` bits would wrongly enable integration at an op), so `analysis("ac")` holds through the whole AC run per LRM 4.6.1 and `@(initial_step("ac"))` fires in AC ([E-53](#)).

#### **dctrcurv.c** (150) + **include/ngspice/trcvdefs.h** (13)

- Generic `.dc @inst[param]` sweeps (a new sweep code): the DC sweep previously hardcoded `Vsource/Isource/Resistor/temperature`, so sweeping any other device parameter was a fatal error. The generic sweep resolves `@inst[param]` through the device's own parameter tables, refreshes per point via `DEVtemperature` (for OSDI exactly the `alter` path), nests with other sweep variables, and restores the original value afterwards ([E-62](#)).
- Fires `final_step` at the last sweep point; honors a `$finish` raised mid-sweep ([E-53](#), [E-55](#)).

#### **noisean.c** (37)

- A singular AC matrix during noise analysis crashed ngspice with a SIGABRT: the code ignored `NIacIter`'s return and the adjoint solve asserted on the unfactored matrix. It now aborts cleanly ("AC solution failed at ... Hz") ([E-56](#)).
- Honors deferred `$finish/$stop` raised at the noise operating point and fires `final_step` on success ([E-55](#), [E-53](#)).

#### **span.c** (31) + **cktspdum.c** (14)

- 1-port S-parameter analyses enabled: the "we need at least two ports" error was over-strict (a 1-port is a plain reflection measurement; the matrix machinery is N-general) — and it was hiding the `cadjoint` crash fixed in `dense.c` ([E-64](#)).
- `Rbase` published into the `sp` plot from port 1's `z0`, making the Touchstone writers work with no manual `let Rbase = 50` incantation ([E-64](#)).
- Stock NaN fixed in `donoise` noise-parameter extraction: the uncorrelated noise conductance `Gu` is analytically zero for fully-correlated single-source topologies, and floating-point rounding could land `sqrt(Ycor2 + Gu/Rn)`'s argument at  $-10^{-18} \rightarrow \text{NaN}$  for the noise figure. Clamped to the physical range  $\geq 0$  — found by parity testing against an OSDI resistor that produced the exact textbook NF where the built-in returned NaN ([E-63](#)).

#### **cktdisto.c** (16)

- `.disto` silently reported zero distortion for OSDI devices — the distortion kernel needs higher-order Taylor coefficients the OSDI ABI cannot provide (first derivatives only), and such devices were skipped without a word. A prominent warning now names each affected OSDI device type ([E-62](#)).

#### 4. Frontend — `src/frontend/`

**plotting/pyplot.c** (new) + **com\_pyplot.c** (new) + **plotit.c, commands.c** (E-94)

A new interactive command, **pyplot**, plots simulated vectors with matplotlib — a Python counterpart to gnuplot. `ft_pyplot()` (`plotting/pyplot.c`) mirrors `ft_gnuplot()`: it writes a `<file>.data` table and a `<file>.py` matplotlib script and `system()`s `python3` (synchronously for a PNG, backgrounded for an interactive window). A new "pyplot" device arm in `plotit()` routes to it, so it accepts the same plot expressions as `plot/gnuplot`; the `com_pyplot()` wrapper mirrors `com_gnuplot()`, and `pyplot` is registered in both command tables. Two set variables tune it: `pyplot_terminal=png` renders headless (Agg) to a PNG, and `pyplot_python` picks the interpreter (default `python3`). Purely additive (E-94). The output file base name is optional (E-95): `com_pyplot()` treats the first word as a file name only when it is not a plot expression (no `(`, and not a vector by `vec_get()`), otherwise it defaults to `pyplot` — so `pyplot v(out)` plots directly (the command's minimum argument count was lowered from 2 to 1). `ft_pyplot()` also grew stacked subplots (set `pyplot_subplots=N` → `plt.subplots(nrows, 1, sharex=True)`, `N` traces per panel; `vs` stays the x-axis, so it is not a panel separator) and matplotlib style sheets (set `pyplot_style=<name>`, `dark` → `dark_background`, guarded) (E-98). The headless `pyplot_terminal` was extended from PNG-only to the vector export formats `svg` and `pdf` (set `pyplot_terminal=svg/pdf` → `fig.savefig(<file>.<fmt>)`, matplotlib picking the writer from the extension), and a figure size control was added (set `pyplot_figsize="W,H"` → `plt.subplots(..., figsize=(W, H))`; quote the value so ngspice keeps the comma) (E-99).

**postcoms.c** (414 diff lines) + **postcoms.h** + **commands.c** (16)

The Touchstone I/O subsystem (E-64, E-72):

- **wrsnp** (with `wrs2p` dispatching to the same handler): Touchstone v1 for any port count — the classic 2-port `S11 S21 S12 S22` column order preserved byte-identically,  $N \geq 3$  in the spec's row-major layout with at most four complex pairs per line;
- writer options `wrsnp <file> [ri|ma|db] [s|y|z] [hz|khz|mhz|ghz]`, any combination in any order: MA/DB formats, Y/Z parameters (normalized to `Rbase`,  $Y \cdot R / Z / R$ , per the v1 spec), frequency units — the option line reflects every choice;
- **rdnsnp**: reads any Touchstone v1 file into a new plot with a Hz frequency scale and complex vectors matching the `.sp` plot's conventions (MA/DB converted back, Y/Z de-normalized, the 2-port column order handled, `Rbase` published so imports round-trip) — measured data diffs against simulation in one `let` expression.

Why: E-63 showed the S-parameter *analysis* was exact but its results could not leave ngspice in the industry-standard format (`wrs2p` demanded a never-created `Rbase` vector), and nothing could bring VNA data in.

**outitf.c** (16)

- Integer operating-point variables recorded per point: `getSpecial` masked with `IF_REAL` only, so an integer opvar saved with `.save @n1[flag]` produced garbage in transient vectors (E-32).
- The plot-memory guard's message printed `%Id` — a Windows-only `printf` length modifier that is undefined behavior elsewhere and rendered the message as the numberless "memory required (Id Bytes)". Now `%zu`: real byte counts on every platform (E-77).

**resource.c** (27)

`ft_ckptspace()` — the "approaching max data size" warning (`RSS > 95%` of `RSS` + free RAM) — now honors `set no_mem_check` (the same opt-out the fatal output-size estimate in `outitf.c` respects; one variable governs both memory guards) and warns once per excursion above the threshold, re-arming below 90%, instead of repeating on every check through a long analysis (E-81).

## **display.c** (1)

#undef NODEV before the local macro definition — the macOS SDK's <sys/param.h> defines an unrelated NODEV ([E-77](#)).

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## 5. Parser and device registry — **src/spicelib/**

### **parser/ingtok.c** (10)

A **.model** card override silently dropped its first parameter when the type name and the opening parenthesis had no space between them (`modname devtype(p1=v1 ...)`): INPgetNetTok's scan did not treat ( as a token boundary, so the type token swallowed the paren and the first parameter name. Found while verifying event-function counters whose model-card overrides mysteriously kept defaults ([E-8](#)).

### **parser/ingmod.c** (5)

`find_model_parameter()` dereferenced `*(device->numModelParms)`, which is NULL for built-ins that take no model cards (VCVS, CCCS, ...) — any *referenced* `.model m vcvs()` card segfaulted, with no OSDI involved at all (an ordinary MOS instance sufficed). This was the true root of the long-documented "module named like a built-in crashes" gotcha. One NULL guard; both shapes now produce clean, located errors ([E-76](#)).

### **devices/dev.c** (51)

- Duplicate device-type registration warned and skipped: `osdi_add_device()` appended every descriptor unconditionally, and the model-card lookup scans front-to-back — so a module name duplicated across two loaded libraries silently resolved to the *first* registration (loading an updated model library gave the stale device with no hint), and a module named like a built-in corrupted model creation. Registration now checks the table case-insensitively and skips duplicates loudly ([E-76](#)).
- **pre\_osdi** path dedup: loading an already-loaded file notes it and skips — with the remedy stated: *"restart ngspice to load a recompiled file"* ([E-76](#), [E-81](#)).

### **osdi/osdiparam.c** (E-93)

Setting a fixed (**localparam**) OSDI parameter from the netlist used to be swallowed silently: `openvaf` exports every `localparam` as a parameter, its "given" flag is forced false, and the written value is overwritten by the parameter-init default — so `.model ... N=8` on a frozen structural width parameter changed nothing, with no feedback. `OSDIparam/OSDIiParam` now test the new `PARAM_FLAG_FIXED` descriptor flag (set by `openvaf`, a free bit of the parameter `flags` field) and emit *"parameter 'N' is a fixed (localparam) value and cannot be set from the netlist; ignored"* instead of silently dropping it ([E-93](#)).

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## 6. Maths — **src/maths/**

**dense/dense.c** (11 substantive lines; the file also carries a whitespace/line-ending normalization from editing)

**cadjoint()** had no  $1 \times 1$  base case: its cofactor loop allocated  $0 \times 0$  and then negative-sized minors, killing ngspice with `malloc: can't allocate -8 bytes` — the crash that had been hiding behind the "at least two ports" guard in `span.c`. The adjugate of `[a]` is `[1]` (the empty minor's determinant is 1 by convention), making `cinverse` of a  $1 \times 1$  equal  $1/a$ ; a  $100\ \Omega$  load on a  $50\ \Omega$  port now writes a proper `.s1p` with  $S_{11} = 1/3$  exactly ([E-64](#)).



### **sparse/spfactor.c** (6)

The Markowitz-product overflow guards compare a double against `LARGEST_LONG_INTEGER` (= `LONG_MAX`), which is not exactly representable as a double; the conversion is now explicit — identical semantics, intentional and warning-free ([E-77](#)).

### **KLU/klu\_multiply.c** (16)

`SMPmultiply`'s KLU path passed `NULL` internal↔external ordering maps to `KLU_matrix_vector_multiply`, which dereferenced them unconditionally (`&NULL[n] = 0x14` for `n = 5`) and segfaulted — so `.option linesearch` crashed under `.option klu`. The line search is the only caller of `SMPmultiply` under KLU, so this path had never been exercised. A `NULL` map now means the identity ordering (`ext = i + 1`, since the KLU CSR is 0-based and the RHS/Solution vectors are 1-based), making the KLU matrix-vector product — and hence the [E-111](#) line-search residual merit — correct under KLU, with a merit sequence numerically identical to Sparse 1.3 ([E-112](#)).

### **KLU/klusmp.c** (SMPcaSolve)

`SMPcaSolve` is the complex adjoint (transposed) solve used by noise and S-parameters. Its Sparse branch calls `spSolveTransposed`, but the KLU branch called the *non-transposed* `klu_z_solve` — silently wrong for any asymmetric matrix (every circuit with a transistor or controlled source), which is why `.noise` was disabled under KLU. It now calls `klu_z_tsolve` (transposed), so KLU noise matches Sparse exactly (including OSDI device models). This is the core of [E-113](#), which also removes the KLU refusal guards in `noisean.c` / `pzan.c` (single-ended pole-zero runs under KLU; balanced-output pole-zero keeps a targeted guard).

### **spicelib/analysis/cktsens.c**

Sensitivity builds an auxiliary perturbation matrix `delta_Y` (holding  $\partial Y / \partial p$ ) via `SMPnewMatrix`, which allocates a plain Sparse 1.3 matrix (`delta_Y->CKTkluMODE = 0`, no `SMPkluMatrix`). It is only *multiplied* against the solution (`SMPmultiply` → `spMultiply`), never factored, and `DEVbindCSC` always binds into `ckt->CKTmatrix`, not `delta_Y` — so it is correctly Sparse in every case. But two KLU-only setup blocks were gated on the main matrix's `CKTkluMODE`, so under `.option klu` they ran and dereferenced the `NULL` `delta_Y->SMPkluMatrix` — a segfault on every DC/AC `.sens` deck. The blocks now gate on `delta_Y->CKTkluMODE` (its own flag, `0`), keeping `delta_Y` Sparse under KLU exactly as it already is under the default solver, while the main `Y` matrix stays KLU. DC and AC sensitivity now match Sparse exactly ([E-114](#)).

### **spicelib/analysis/distoan.c**

Distortion is a complex analysis (Volterra series solved at the harmonic / intermodulation frequencies via `NIIdIter` → `SMPcSolve`), but `distoan.c` had no KLU code at all: under `.option klu` the KLU matrix stayed in *real* mode, the complex solves ran against an unconverted matrix, and every harmonic came back zero — a silent wrong answer. The fix mirrors `acan.c`: before the solve loop it converts the matrix to complex (`DEVbindCSCComplex` per device + `KLUmatrixIsComplex`), and on exit converts back to real (`DEVbindCSCComplexToReal`) so a later real analysis in the same session is unaffected. KLU distortion now matches Sparse bit-for-bit (single-tone, two-tone intermodulation, and OSDI models) ([E-115](#)).

## 7. Build system

### xspice/icm/makedefs.in (4)

The XSPICE codemodel (.cm) link rule hardcoded the pre-macOS-10.3 idiom `-bundle -flat_namespace -undefined suppress`, deprecated loudly by the modern linker on all 14 codemodel links (CI macOS builds included). Now `-bundle -undefined dynamic_lookup` — the modern spelling for plugins whose host symbols resolve at load time (E-77).

### Autotools regeneration (the ~100 **Makefile.in** files + **config.h.in**)

Regenerated by `./autogen.sh` with libtool 2.5.4 during the zero-warning work — a build tree whose configure predates libtool 2.4.7 selects the deprecated darwin `-undefined suppress` flag whenever `MACOSX_DEPLOYMENT_TARGET` is unset. No hand-written content (E-77).

## 8. Per-enhancement index

Every enhancement that touched ngspice, oldest first:

| Enhancement | ngspice files                                                                                         | One line                                                              |
|-------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| E-1         | osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c, osdisetup.c, dctran.c (+accept path)       | absdelay(): synthetic delay rows + waveform history                   |
| E-6         | osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c                                             | last_crossing(): history + interpolated crossings                     |
| E-8         | parser/ingptok.c                                                                                      | .model card first-parameter drop fix                                  |
| E-24        | osditrunc.c                                                                                           | \$discontinuity next-step clamp                                       |
| E-25        | osdi callbacks (get_simparams)                                                                        | expose analysis_name + simulator simparams                            |
| E-32        | outitf.c                                                                                              | integer opvars recorded per point                                     |
| E-42        | osdinoise.c                                                                                           | coherent same-named noise grouping (LRM 4.6.4)                        |
| E-45        | osdi.h (ABI 0.5), osdisetup.c                                                                         | Verilog-A nodesets applied to the solver                              |
| E-51        | osdi.h (ABI 0.6), osdiacld.c                                                                          | full ac_stim AC-RHS injection                                         |
| E-53        | dctran.c, dcop.c, dctrcurv.c, acan.c, noisean.c, osdiload.c                                           | @(final_step) + analysis-name phase bits                              |
| E-54        | osdi.h (ABI 0.7), osdinoise.c, osdiload.c, osdiacld.c, osdiregistry.c                                 | node-free complex noise factors, stride-2 pairs                       |
| E-55        | osdiaccept.c (new), osdiload.c, osditrunc.c, osdiinit.c, dctran.c, dctrcurv.c, osdiitf.h, Makefile.am | \$finish/\$stop/\$fatal honored; \$discontinuity step rejection       |
| E-56        | noisean.c, osdisetup.c                                                                                | noise SIGABRT fix; setup-rejection diagnostics                        |
| E-62        | dctrcurv.c, trcvdefs.h, cktdisto.c                                                                    | generic .dc @inst[param] sweeps; .disto warning                       |
| E-63        | span.c                                                                                                | donoise NaN clamp (stock defect, found by parity)                     |
| E-64        | span.c, cktspdum.c, dense.c, postcoms.c/.h, commands.c                                                | Touchstone export, auto-Rbase, 1-port .sp, cadjoint 1x1               |
| E-72        | postcoms.c/.h, commands.c                                                                             | Touchstone round 2: MA/DB/Y/Z/units + rdsnp reader                    |
| E-76        | dev.c, inpgmod.c                                                                                      | duplicate-registration warning, load dedup, stock .model segfault fix |

| Enhancement | mspice files                                                                           | One line                                                                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-77        | osddefs.h, display.c, outitf.c, spfactor.c, makedefs.in (+autotools regen)             | zero-warning build, 33 → 0                                                                                                                                                                                                                                                        |
| E-80        | osdiinit.c                                                                             | dtemp instance-parameter alias                                                                                                                                                                                                                                                    |
| E-81        | resource.c, dev.c                                                                      | memory-warning opt-out + once-per-excursion; reload-note hint                                                                                                                                                                                                                     |
| E-93        | osdi.h, osdiparam.c                                                                    | warn (not silently ignore) when a netlist sets a PARA_FLAG_FIXED (localparam) OSDI parameter                                                                                                                                                                                      |
| E-94        | plotting/pyplot.c (new), com_pyplot.c (new), plotit.c, commands.c                      | new pyplot command — plot vectors with matplotlib (a Python gnuplot)                                                                                                                                                                                                              |
| E-95        | com_pyplot.c, commands.c                                                               | make the pyplot output file name optional (defaults to pyplot)                                                                                                                                                                                                                    |
| E-98        | plotting/pyplot.c                                                                      | pyplot stacked subplots (pyplot_subplots=N) + matplotlib style sheets (pyplot_style)                                                                                                                                                                                              |
| E-99        | plotting/pyplot.c                                                                      | pyplot vector export formats (pyplot_terminal=svg/pdf) + figure size (pyplot_figsize)                                                                                                                                                                                             |
| E-110       | optdefs.h, tsdefs.h, cktsopt.c, cktntask.c                                             | .option errpreset=conservative moderate liberal — one knob for a coordinated tolerance/robustness set; explicit options override regardless of order (moderate = historical defaults)                                                                                             |
| E-111       | optdefs.h, tsdefs.h, cktdefs.h, cktsopt.c, cktdojob.c, cktntask.c, cktdest.c, niiter.c | .option linesearch — globalized (damped) Newton via Armijo backtracking on a new KCL-residual merit<br>□ F□ =□ G·x-b□ ; result-neutral, off by default                                                                                                                            |
| E-112       | maths/KLU/klu_multiply.c                                                               | KLU support for .option linesearch — SMPmultiply's KLU path passed NULL ordering maps that klu_matrix_vector_multiply dereferenced (SIGSEGV); NULL now means identity ordering. Line search verified merit-identical KLU vs Sparse                                                |
| E-113       | maths/KLU/klusmp.c, spicelib/analysis/noisean.c, pzan.c                                | KLU support for noise + single-ended pole-zero — SMPcaSolve's adjoint KLU branch used the non-transposed solve (silently wrong noise on asymmetric matrices); now klu_z_tsolve. Guards removed; balanced-output pz keeps a targeted guard                                         |
| E-114       | spicelib/analysis/cktsens.c                                                            | KLU support for sensitivity — the auxiliary perturbation matrix delta_Y is Sparse, but two KLU setup blocks gated on the main matrix's flag dereferenced its NULL SMPkluMatrix (segfault on every DC/AC .sens); now gated on delta_Y's own flag. DC/AC .sens match Sparse exactly |
| E-115       | spicelib/analysis/distoan.c                                                            | KLU support for distortion — .disto is a complex analysis but distoan.c had no KLU code, so under KLU the matrix stayed real and every harmonic came back zero (silent wrong answer); now converts real↔complex around the solve loop like acan.c. Matches Sparse bit-for-bit     |
| E-116       | osdi/osdisetup.c                                                                       | KLU wrong-DC fix — an OSDI internal node with no Jacobian entry (a decoupled ground reference) was given an all-zero solver row that made the KLU matrix structurally singular; now tied to ground. Fixes the groundcontrib and hierbranch KLU_XFAILs                             |

| Enhancement | ngspice files                                                                                                         | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-117       | configure.ac (+configure),<br>spicelib/analysis/dcpss.c                                                               | Periodic steady state (PSS) productionized — built by default (was experimental <code>--enable-pss</code> , so <code>.pss</code> was unimplemented in shipped builds); ~230 lines of shooting-loop trace routed through <code>set ngdebug</code> (232 → 31 lines by default); fail-fast KLU guard (PSS hangs under KLU) directing <code>.pss</code> to <code>.option sparse</code>                                                                                                                                                                                                                                                                                                           |
| E-118       | spicelib/analysis/dcpss.c                                                                                             | PSS runs under KLU — the shooting loop hung under KLU ( <code>klu_refactor</code> 's reused pivots inflated the truncation error into a ~20M-step timestep explosion); forcing a full re-factor ( <code>NISHOULDREORDER</code> ) each PSS step makes KLU converge to the same result as Sparse. E-117's Sparse-only guard removed                                                                                                                                                                                                                                                                                                                                                            |
| E-119       | pssdefs.h,<br>spicelib/analysis/dcpss.c                                                                               | Retain the PSS periodic operating point — PSS sampled the node voltages per period for its DFT then freed them, and never captured device states; now the voltages and <code>CKTstate0</code> states are captured per sample and retained on the PSSan job (with frequency + dims) as the substrate for the periodic small-signal suite (PAC/pnoise/PXF). Self-checks that the retained samples reproduce the fundamental                                                                                                                                                                                                                                                                    |
| E-120       | spicelib/analysis/dcpss.c                                                                                             | Periodic small-signal Jacobian harmonics — walk the retained operating point, re-linearize each sample ( <code>CKTload MODEINITSMSIG</code> ) and stamp $G+jC$ ( <code>CKTacLoad <math>\omega=1</math></code> ), read the osc-node diagonal → $g(t)$ , $c(t)$ , DFT to harmonics. The blocks the PAC conversion matrix is built from. (Fixed a complex-mode-not-set bug where $C(t)$ accumulated across samples.) Verified: RC gives $G=1/R1$ , $C=C1$ , no harmonics                                                                                                                                                                                                                        |
| E-121       | spicelib/analysis/dcpss.c                                                                                             | PAC conversion-matrix engine — extend E-120 from the osc diagonal to every matrix nonzero, complex-DFT to harmonics $G_k$ , $C_k$ , assemble the $(2M+1)N$ harmonic conversion matrix $H_{\{nm\}}=G_{\{n-m\}}+j\omega_m \cdot C_{\{n-m\}}$ , inject a unit current at the osc node in sideband 0 and solve by dense complex LU ( <code>pss_csolve</code> ), report the sideband conversion gains. The numerical heart of PAC/pnoise/PXF. Verified: linear RC gives sideband-0 = AC driving-point                                                                                                                                                                                             |
| E-122       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | <code>.pac</code> command (periodic AC) — the user-facing analysis on the E-121 engine. <code>.pac Fguess StabTime OscNode Points Harmonics SC_iter Steady_coeff</code><br><dec oct lin> Npts Fstart Fstop runs PSS then sweeps the input frequency, solving the conversion matrix at each point and emitting the 0-th-sideband node responses as a complex PAC Analysis plot ( <code>print/plot/wrdata</code> ). Reuses the PSS analysis via a <code>PSSdoPAC</code> flag; engine factored into <code>pac_extract_harmonics</code> (once) + <code>pac_solve_at</code> (per freq). Verified: swept sideband-0 $ b(f)  ==$ analytic AC driving-point $ Z(f) $ to $1.6e-7$ across 10 kHz–1 MHz |

| Enhancement | ngspice files                                                                                                         | One line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-123       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | finish .pac — (1) source-referenced stimulus: capture the netlist AC-source RHS from CKTacLoad as the sideband-0 stimulus B_0 (a periodic-AC transfer/conversion gain), unit-current-at-osc-node fallback; (2) multi-sideband output: optional trailing maxsideband Ksb emits every conversion sideband $f_{in}+k \cdot f_0$ ( $k=-Ksb \dots Ksb$ ) as its own vector — sideband 0 keeps the node name, others <code>&lt;node&gt;_usb&lt;k&gt;/&lt;node&gt;_lsb&lt;k&gt;</code> via IFnewUId. Verified: RC with V1 AC 1 gives sideband-0 = low-pass transfer $1/\sqrt{1+(2\pi fRC)^2}$ (0.998→0.157) with $b_{usb1}/b_{lsb1} \sim 0$ (no conversion)                                                                                                                 |
| E-124       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | .pnoise command (periodic noise) — fold each device's noise through the conversion matrix. <code>pac_solve_adjoint</code> solves $H^T \Psi = e_{\{out,0\}}$ ; <code>pnoise_sweep</code> loads the sideband-k adjoint into CKTrhs/CKTirhs and calls the existing device noise routines (NevalSrc, OSDI <code>load_noise</code> ) once per sideband, accumulating $\sum_k S \cdot  \Delta \Psi_k ^2$ — reuses every device noise model via a minimal local NOISEAN context. <code>.pnoise &lt;pss&gt; OutNode InSrc &lt;dec oct lin&gt; Np Fstart Fstop</code> ; emits <code>onoise_spectrum/inoise_spectrum</code> . Verified: linear RC reduces exactly to <code>.noise</code> ( $4kTR/(1+(2\pi fRC)^2)$ ), matches the <code>.noise</code> reference to every digit |
| E-125       | include/ngspice/pssdefs.h,<br>spicelib/analysis/psssetp.c,<br>spicelib/parser/inp2dot.c,<br>spicelib/analysis/dcpss.c | .pxf command (periodic transfer function) — the adjoint of PAC, completing the PSS→PAC→Pnoise→PXF suite. <code>pxf_sweep</code> solves $H^T \Psi = e_{\{out,0\}}$ per frequency and dots each sideband block with the AC-source pattern $B_0 \rightarrow x_{f_k} = \sum_j \Psi_k(j) \cdot B_0(j)$ , the input→output transfer at each sideband. <code>.pxf &lt;pss&gt; OutNode &lt;dec oct lin&gt; Np Fstart Fstop [maxsb]</code> ; emits <code>xf + xf_usb&lt;k&gt;/xf_lsb&lt;k&gt;</code> . Verified: by the identity $(H^{-1}B)_{out} = (H^{-T}e_{out})^T B$ the sideband-0 transfer is bit-identical to the PAC response (0.998→0.157 low-pass), conversion sidebands $\sim 2e-16$                                                                               |

(E-25's *simpam* exposure lives in the OSDI callback table populated at load time; its diff rides inside the *osdi\_load.c/callbacks* changes.)

Every entry above is guarded by a verify script under [examples/](#) and was regression-checked against the full example arsenal, the compiler's integration suite, and the VA\_TEST industry-model corpus at the time it landed.